

Notes: Davis, Haltiwanger, Handley, Jarmin, Lerner, and Miranda (AER, 2014)

Private Equity, Jobs, and Productivity

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1 Big picture

- **Question.** What happens to employment, job reallocation, productivity, and worker earnings after private equity buyouts?
- **Motivation.** Critics of private equity claim that buyouts mainly cut jobs and burden firms with debt, while industry defenders claim that private equity improves operations and growth. The paper brings establishment-level data to this debate.
- **Main data contribution.** The authors construct a new matched data set linking roughly 3,200 U.S. firms acquired in private equity buyouts from 1980 to 2005 to the Census Bureau’s Longitudinal Business Database, and then track approximately 150,000 target establishments.
- **Key measurement advantage.** Establishment-level data separate organic employment changes from acquisitions, divestitures, births, deaths, and within-firm restructuring. This is essential because firm-level employment can change even when jobs at underlying establishments do not.
- **Main employment result.** Employment at target establishments grows more slowly than employment at matched controls after buyouts. The cumulative two-year difference is modest but negative.
- **Main reallocation result.** Buyouts sharply increase both job creation and job destruction. Net job losses are modest, but gross job flows are large.
- **Main firm-level result.** At the firm level, the picture is more nuanced: target firms shrink relative to controls on continuing establishments but create more jobs through new establishments and through acquisitions.
- **Main productivity result.** In manufacturing, target firms improve total factor productivity relative to controls. The productivity gains come mainly from reallocating activity away from low-productivity plants and toward higher-productivity entrants.
- **Main earnings result.** Earnings per worker fall at target establishments relative to controls after buyouts, suggesting a shift away from high-EPW establishments or toward lower labour costs.
- **Takeaway.** Private equity buyouts are not simply job-destruction events. They are restructuring events that accelerate creative destruction: more job creation, more job destruction, higher productivity, and some reduction in earnings per worker.

2 Incremental contribution of the paper

2.1 Contribution relative to prior private equity studies

- Prior studies often use small samples, survey data, or public-to-private buyouts from the 1980s.
- Many prior analyses observe firm-level employment only. This makes it hard to tell whether employment changes reflect true organic employment change or whether they arise because a firm acquires or sells establishments.
- Davis et al. exploit comprehensive Census Bureau establishment-level data to observe jobs at individual factories, offices, stores, and other physical locations before and after a buyout.
- The paper’s scope is much larger than earlier U.S. studies: it covers buyouts from 1980 through 2005 and includes the massive expansion of private equity activity after the mid-1990s.

Interpretation: The paper shifts the debate from anecdotal claims and narrow samples to large-scale, matched establishment evidence.

2.2 Contribution relative to job-flow research

- Labour-market research emphasizes that net employment changes hide large underlying flows of job creation and job destruction.
- The paper applies this job-flow perspective to private equity buyouts.
- It shows that buyouts increase reallocation even when net employment effects are modest.
- This means that private equity's impact is not well summarized by a single employment-growth coefficient.

Interpretation: The paper connects corporate finance to the job-flow literature by treating buyouts as catalysts for reallocation across establishments and within firms.

2.3 Contribution relative to productivity studies

- Prior productivity work often focuses on manufacturing plants or case studies.
- This paper links buyout information to plant-level manufacturing productivity measures from the Census of Manufactures and Annual Survey of Manufactures.
- It decomposes productivity gains into continuing-plant productivity, exit, entry, acquisitions, and divestitures.
- The key result is that productivity improvements primarily reflect selection and reallocation rather than productivity growth within continuing plants.

Interpretation: The paper contributes an explicit micro-decomposition of private equity's productivity effects.

2.4 Contribution relative to policy debate

- The public debate often frames private equity as either value-destroying job cutting or value-creating operational improvement.
- The paper provides a more nuanced empirical account: buyouts lead to modest net job losses, substantial job reallocation, and productivity gains.
- This nuance matters for evaluating labour-market effects, productivity effects, and the welfare consequences of buyouts.

Interpretation: The incremental value of the paper is that it gives a measured, multi-margin answer rather than a one-dimensional answer.

3 Data and measurement

3.1 Private equity transaction data

- The authors identify U.S. private equity buyouts from Capital IQ and other sources.
- The matched sample includes 3,218 buyout transactions from 1980 to 2005.
- Transactions are classified into private-to-private, public-to-private, divisional sales, secondary sales, and other deal types.
- The paper focuses on buyouts, not venture capital-backed firms.

Interpretation: Distinguishing buyout types matters because public-to-private deals, private-to-private deals, divisional sales, and secondary buyouts can have very different employment effects.

3.2 Census Bureau establishment and firm data

- The Longitudinal Business Database covers almost all U.S. nonfarm private-sector employers.
- It contains annual establishment-level employment, payroll, industry, location, ownership links, and identifiers.
- The data make it possible to follow target establishments and matched controls before and after the buyout.
- They also allow the authors to construct firm-level outcomes by aggregating establishments and tracking births, deaths, acquisitions, and divestitures.

Interpretation: Establishment data are essential because private equity firms often restructure by buying, selling, opening, or closing establishments. Firm-level data alone would confound these margins.

3.3 Manufacturing productivity data

- The productivity analysis uses the Census of Manufactures and Annual Survey of Manufactures.
- Plant-level TFP is measured using output, capital, labour, materials, energy, and industry-level factor shares.
- The manufacturing productivity sample is smaller than the employment sample but allows the paper to study operating efficiency.

Interpretation: Manufacturing is the only sector where plant-level inputs and outputs are rich enough for direct TFP analysis.

3.4 Matched controls

The paper builds matched controls using cells based on:

- industry;
- firm size category;
- firm age category;
- single- versus multi-establishment status;
- transaction year;
- pre-buyout employment growth history.

Interpretation: The goal is to compare target firms and establishments to plausible counterfactuals facing similar industry-year and life-cycle conditions.

4 Empirical specifications and core definitions

4.1 Employment growth measure

The paper uses the standard Davis-Haltiwanger-Schuh growth-rate measure:

$$g_{it} = \frac{E_{it} - E_{i,t-1}}{X_{it}}, \quad X_{it} = \frac{1}{2}(E_{it} + E_{i,t-1}), \quad (1)$$

where E_{it} is employment at establishment or firm i in year t .

Interpretation: This growth rate behaves like a log difference for continuing units but can accommodate establishment entry and exit because it is bounded between -2 and 2 .

4.2 Nonparametric matched comparisons

For each event year relative to the buyout, the paper compares target outcomes with control outcomes within matched cells:

$$\Delta_\tau = E[g_{i,t+\tau} \mid Target] - E[g_{i,t+\tau} \mid Control]. \quad (2)$$

Interpretation: This event-study comparison shows the evolution of target-control differences before and after the buyout without imposing a parametric regression structure.

4.3 Semi-parametric regressions

The semi-parametric specifications estimate target-control differences while controlling for a rich set of interactions:

$$y_{it} = \alpha + \beta Target_i + \Gamma' Controls_{it} + \varepsilon_{it}. \quad (3)$$

The controls include a full cross-product of industry, year, firm size, firm age, multi-unit status, and pre-buyout growth controls.

Interpretation: The coefficient on $Target_i$ estimates the average target-control difference conditional on rich observable characteristics.

4.4 Firm-level adjustment margins

Firm-level employment changes are decomposed into:

- continuing-establishment growth;
- job creation at continuing establishments;
- job destruction at continuing establishments;
- establishment births;
- establishment deaths;
- acquisitions;
- divestitures.

Interpretation: This decomposition is where the paper shows that private equity buyouts trigger creative destruction: greater births and acquisitions as well as greater deaths and divestitures.

4.5 Productivity decomposition

The manufacturing productivity analysis decomposes target-control TFP differences into continuing plants, entry and exit, acquisitions and divestitures:

$$\Delta P_t - \tilde{\Delta P}_t = \text{continuers} + \text{entry/exit} + \text{acquisitions/divestitures}. \quad (4)$$

Interpretation: The decomposition asks whether productivity gains come from improving the same plants or from reallocating activity toward more productive plants.

5 Identification and interpretation

5.1 Identification logic

- The paper is observational rather than experimental.
- Identification relies on comparing buyout targets to control firms and establishments matched on rich observables.
- Conditional mean independence is the key assumption: conditional on controls and treatment status, target and control counterfactual outcomes are comparable.
- The very large Census microdata allow unusually detailed matching relative to previous studies.

Interpretation: The empirical design is best understood as a rich matched-control event study, not as a randomized design.

5.2 What the estimates identify

- Establishment-level estimates identify organic changes at establishments owned by the target firm at the time of buyout.
- Firm-level estimates identify broader changes that include births, deaths, acquisitions, and divestitures.
- Manufacturing estimates identify changes in productivity for plants observed in the manufacturing data.

Interpretation: The paper carefully separates these layers because each answers a different economic question.

5.3 Limitations

- Private equity targets may still differ from controls on unobserved dimensions.
- Firm tracking becomes difficult after complex restructurings, which is why the firm-level window is limited to two years.
- Productivity is directly measured only in manufacturing.
- The paper does not observe every contractual or governance mechanism used by private equity owners.

Interpretation: The strongest evidence comes from the consistency of multiple margins: establishment employment, firm job flows, productivity, and earnings per worker.

6 Tables and figures

6.1 Tables 1–2 and Figures 1–2: buyout sample and time series

Read:

- Table 1 reports the transaction value of private equity buyout targets by decade.
- Table 2 describes the analysis samples, including 3,218 transactions from 1980–2005 and 151,529 target establishments.
- Figure 1 shows the rising number of U.S. private equity buyout targets, especially after the mid-1990s.
- Figure 2 shows target employment as a share of LBD employment and in thousands.

Economic message: The paper studies a much broader and more modern private equity universe than earlier work, and the matched sample tracks a large share of buyout activity.

TABLE 1—VALUE OF PRIVATE EQUITY BUYOUT TARGETS, TOTAL AND MATCHED BY DECADE
(\$millions)

	1980–1989	1990–1999	2000–2005
Matched	138,111	212,387	363,395
Total	199,166	280,746	529,595

Note: Enterprise values (debt plus equity) as of the transaction are imputed for about half the buyout targets based on deal type, industry, and transaction year, as detailed in online Appendix A.

TABLE 2—PRIVATE EQUITY BUYOUTS IN THE ANALYSIS SAMPLES

	Number of transactions (target firms)	Value of transactions, millions of dollars	Target establishments, transaction year	Target employment, transaction year
1980–2005	3,218	713,892	151,529	5,828,532
Private to private	1,350	149,509	59,865	2,224,530
Public to private	390	272,599	36,717	1,371,129
Divisional sales	918	158,580	35,259	1,359,139
Secondary sales	396	108,324	13,455	637,591
Other	164	24,880	6,233	236,143
1980–2003	2,265	530,786	103,671	4,323,558
Excluding EIN cases	1,874	465,657	79,131	3,410,598
Two-year continuers, excluding EIN cases	1,374	333,519	76,271	3,187,171
Private to private	686	91,247	37,283	1,470,447
Public to private	248	133,727	20,380	872,206
Divisional sales	206	45,683	7,922	391,705
Secondary sales	160	47,627	7,957	353,325
Other	74	15,235	2,729	99,488
1980–2000	1,306	395,020	54,729	2,385,163
Private to private	647	99,973	24,593	901,284
Public to private	171	173,500	18,454	854,779
Divisional sales	342	74,487	6,557	416,055
Secondary sales	107	35,742	3,885	161,557
Other	39	11,318	1,240	51,488
g., 1980–2003	539	138,006	9,174	805,328
Multi-unit firms only	427	126,572	9,062	792,864
Multi-unit with TFP	286	82,076	2,053	496,699

tes: We exclude single-unit matched targets from our analysis of the manufacturing sector. They account for only 2 of 9,744 target manufacturing establishments and less than 1.5 percent of target manufacturing employment. e last row in the table reports data for matched multi-unit targets for which we can obtain data on total factor productivity in the transaction year from the Census of Manufactures or the Annual Survey of Manufactures.

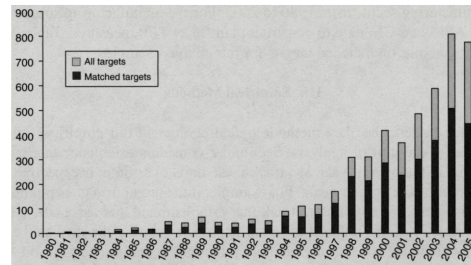


FIGURE 1. NUMBER OF TARGET FIRMS IN US PRIVATE EQUITY BUYOUTS, 1980 TO 2005

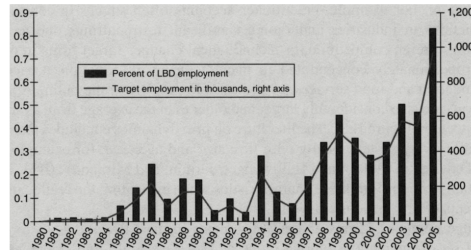


FIGURE 2. EMPLOYMENT AT MATCHED TARGETS AS OF THE BUYOUT YEAR, 1980 TO 2005

Tables 1–2: sample overview

Figures 1–2: buyout activity over time

6.2 Figures 3–5 and Table 3: establishment-level employment effects

Read:

- Figure 3A compares employment trajectories at target and control establishments.
- Figure 3B shows target-minus-control employment growth differences around the buyout year.
- Figures 4A and 4B show that post-buyout employment effects come from both job creation and job destruction margins.
- Figure 5 shows that target establishments have higher post-buyout exit rates.
- Table 3 confirms in regressions that target establishments grow more slowly after buyouts, especially in years $t + 1$, $t + 2$, $t + 4$, and $t + 5$.

Economic message: Establishments owned by target firms experience modestly lower employment growth after buyouts, driven mainly by higher job destruction and exit.

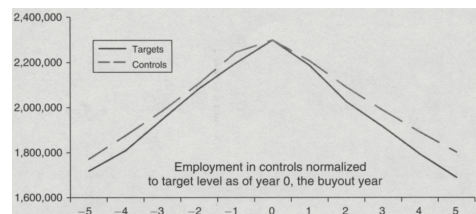


FIGURE 3A. COMPARISON OF EMPLOYMENT TRAJECTORY FOR TARGET ESTABLISHMENTS TO CONTROLS, BUYOUTS FROM 1980 TO 2000

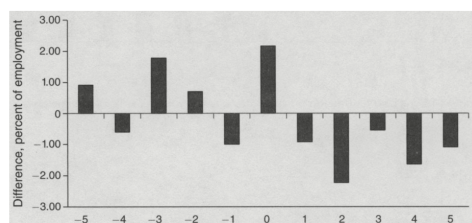


FIGURE 3B. EMPLOYMENT GROWTH RATE DIFFERENCES BEFORE AND AFTER THE BUYOUT YEAR, TARGET ESTABLISHMENTS MINUS CONTROLS, BUYOUTS FROM 1980 TO 2000

Figure 3: employment trajectory and growth differences

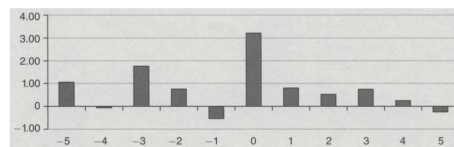


FIGURE 4A. JOB CREATION RATES BEFORE AND AFTER BUYOUT YEAR, TARGET ESTABLISHMENTS MINUS CONTROLS, BUYOUTS FROM 1980 TO 2000

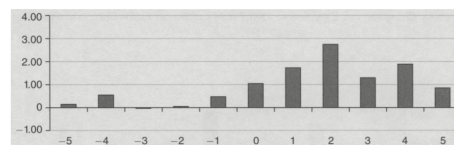


FIGURE 4B. JOB DESTRUCTION RATES BEFORE AND AFTER BUYOUT YEAR, TARGET ESTABLISHMENTS MINUS CONTROLS, BUYOUTS FROM 1980 TO 2000

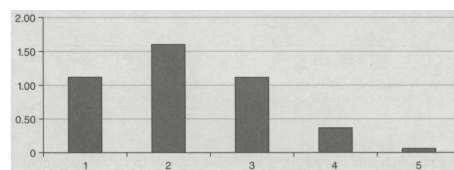


FIGURE 5. EMPLOYMENT-WEIGHTED ESTABLISHMENT EXIT RATES POST BUYOUT, TARGETS MINUS CONTROLS, BUYOUTS FROM 1980 TO 2000

Figures 4–5: job flows and exit

TABLE 3—POST-BUYOUT EMPLOYMENT GROWTH RATES AT TARGET ESTABLISHMENTS RELATIVE TO CONTROLS, BUYOUTS FROM 1980 TO 2000

	Nonparametric comparison from Figure 3B	Semi-parametric regressions	
		ATE=ATE1	ATE1 heterogeneous
Buyout year	2.17	2.08*** (0.17)	2.28*** (0.17)
Buyout year +1	-0.93	-0.72*** (0.20)	-1.15*** (0.20)
+2	-2.23	-1.74*** (0.20)	-1.76*** (0.21)
+3	-0.55	0.00 (0.21)	0.08 (0.21)
+4	-1.64	-1.31*** (0.22)	-1.16*** (0.22)
+5	-1.09	-0.95*** (0.22)	-1.23*** (0.23)

Notes: Table entries report estimated employment growth rate differences between targets and controls in the buyout year and following years. For example, the entries for “Buyout year +2” report the estimated growth rate difference from Year 1 to Year 2 following the buyout. Each reported coefficient is for a different nonparametric comparison or regression. Standard errors are in parentheses. They are computed by the delta method in the “ATE1 heterogeneous” regression. The nonparametric comparison reflects the patterns displayed in Figure 3b. As explained in the text, this comparison allows for heterogeneous treatment effects and controls for the fully interacted cross product of 72 industries, 10 firm size categories, 6 firm age groups, multi-unit status, and buyout year. The semi-parametric regression specifications include fully interacted industry, year, firm age, firm size, and multi-unit effects plus additional controls for pre-buyout growth history. The “ATE=ATE1” specification imposes a uniform treatment effect, while the “ATE1 heterogeneous” specification allows the treatment effect to vary by firm size, firm age, and the two measures of pre-buyout growth history. The average number of establishment-level observations in each regression or nonparametric comparison is about 4.9 million. The observation count falls with each successive year following the transaction year because of target deaths and deleted observations for the corresponding control establishments.

***Significant at the 1 percent level.

Table 3: establishment employment growth relative to controls

6.3 Tables 4–6: firm-level margins and job reallocation

Read:

- Table 4 decomposes firm-level employment growth into continuers, creation, destruction, deaths, births, acquisitions, and divestitures.

- Target firms shrink relative to controls on continuing establishments but create more jobs through births and acquisitions.
- Table 5 shows that buyouts substantially increase excess reallocation and job reallocation at target firms.
- Table 6 aggregates job creation, destruction, employment growth, job reallocation, and excess reallocation for target and control firms.

Economic message: Private equity buyouts accelerate restructuring. Net employment effects are modest, but gross job flows are large.

TABLE 4—BUYOUT EFFECTS ON EMPLOYMENT GROWTH RATE AT TARGET FIRMS RELATIVE TO CONTROLS, BUYOUTS FROM 1980 TO 2003

dependent variable:	Regression specification			
	ATE=ATE1		ATE1 heterogeneous	
	Buyout effect	R ²	Buyout effect	R ²
firm-level employment growth rate from buyout year t to $t + 2$	-0.88 (0.18)	0.07	-0.65 (0.16)	0.07
γ adjustment margin				
Continuers	-1.57 (0.12)	0.09	-1.36 (0.11)	0.09
Creation	-1.07 (0.08)	0.16	-0.93 (0.08)	0.16
Destruction	0.71 (0.07)	0.09	0.64 (0.07)	0.09
Deaths	4.12 (0.09)	0.06	4.13 (0.08)	0.06
Births	1.80 (0.05)	0.22	1.87 (0.05)	0.22
Acquisitions	5.62 (0.05)	0.12	5.56 (0.05)	0.13
Divestitures	2.77 (0.05)	0.06	2.75 (0.04)	0.06

Notes: Employment-weighted regressions on target and control firms, with rates calculated over the two-year horizon from the event year t to $t + 2$. Standard errors in parentheses. All reported coefficient values are statistically significant at the 0.01 level. The semi-parametric regression specifications include fully interacted industry, year, firm age, firm size, and multi-unit effects plus additional controls for pre-buyout growth history. The "ATE=ATE1" specification imposes a uniform treatment effect, while the "ATE1 heterogeneous" specification allows the treatment effect to vary by firm size, firm age, and the two measures of pre-buyout growth history. Each regression has 985,000 (rounded to nearest 1,000) observations.

pecification. These estimated effects are much smaller than the cumulative two-year difference of 2.9 points for both specifications in Table 3. This comparison suggests

Table 4: firm-level employment margins

TABLE 5—BUYOUT EFFECTS ON FIRM-LEVEL JOB REALLOCATION AND EXCESS REALLOCATION, BUYOUTS FROM 1980 TO 2003

dependent variable:	Regression specification			
	ATE=ATE1		ATE1 heterogeneous	
	Buyout effect	R ²	Buyout effect	R ²
estimated responses relative to controls from buyout year t to $t + 2$				
firm-level excess reallocation—All adjustment margins	9.25 (0.08)	0.37	9.29 (0.09)	0.37
firm-level excess reallocation—Births, deaths, and continuers	6.38 (0.08)	0.38	6.40 (0.08)	0.39
firm-level job reallocation—All adjustment margins	13.81 (0.15)	0.21	13.89 (0.15)	0.21
firm-level job reallocation—Births, deaths, and continuers	5.47 (0.14)	0.22	5.62 (0.14)	0.22

Notes: Employment-weighted regressions on a sample of target and control firms, with rates calculated over the two-year horizon from the event year t to $t + 2$. Standard errors in parentheses. All reported coefficient values are statistically significant at the 0.01 level. The semi-parametric regression specifications include fully interacted industry, year, firm age, firm size, and multi-unit effects plus additional controls for pre-buyout growth history. The

Table 5: reallocation and excess reallocation

TABLE 6—CUMULATIVE TWO-YEAR JOB REALLOCATION AT TARGET FIRMS AND CONTROLS, BUYOUTS FROM 1980 TO 2003

ates expressed as a percent employment	Target firms	Control firms	Difference	From Tables 4 and 5	
				Difference	Standard error
<i>mel A. Organic changes, excluding acquisitions and divestitures</i>					
b creation	24.96	22.96	2.00	0.94	(0.10)
Continuers	11.51	11.74	-0.22	-0.93	(0.07)
Births (entrants)	13.44	11.22	2.22	1.87	(0.05)
b destruction	26.89	22.62	4.27	4.69	(0.11)
Continuers	13.28	12.65	0.63	0.64	(0.06)
Deaths (exits)	13.60	9.96	3.64	4.13	(0.08)
employment growth	-1.93	0.34	-2.27	-3.75	(0.16)
b reallocation	51.85	45.58	6.27	5.62	(0.14)
cess reallocation	49.91	45.23	4.68		
Within-firm	33.09	27.02	6.06	6.40	(0.08)
<i>mel B. All adjustment margins, including acquisitions and divestitures</i>					
b creation	35.86	28.42	7.44	6.25	(0.11)
b destruction	33.21	26.67	6.53	7.03	(0.13)
employment growth	2.65	1.75	0.90	-0.65	(0.16)
b reallocation	69.07	55.10	13.97	13.89	(0.15)
cess reallocation	66.41	53.35	13.07		
Within-firm	38.82	29.82	9.01	9.29	(0.09)

Notes: For Target firms and Control firms, we aggregate over cells using the employment shares of targets. For cells with multiple controls, each control receives equal weight.

equity groups shifted to a more operational orientation over time, which could lead to time-varying effects on target employment. The scale of private buyout activity also increased enormously over time, which could alter the character of the marginal

Table 6: cumulative two-year job reallocation

6.4 Table 7: heterogeneity by buyout type

Read:

- Public-to-private deals have large negative employment growth relative to controls.

- Independent-to-private deals have positive employment growth relative to controls.
- Excess reallocation is positive for all buyout types.

Economic message: The employment effects of private equity differ sharply by deal type, but the reallocation effect is broad-based.

TABLE 7—BUYOUT EFFECTS ON TARGET FIRMS RELATIVE TO CONTROLS BY TYPE OF BUYOUT, 1980 TO 2003

Dependent variable:	Type of private equity buyout				
	Public to private	Independent to private	Divisional buyout	Secondary buyout	Other
Employment growth rate on buyout year t to $t + 2$	-10.36 (0.38)	10.51 (0.24)	-1.48 (0.45)	7.15 (0.58)	-6.45 (0.80)
Excess reallocation rate from buyout year t to $t + 2$	5.08 (0.21)	4.69 (0.10)	20.32 (0.19)	29.79 (0.27)	6.16 (0.40)
Observations (rounded)	374,000	1,298,000	475,000	169,000	123,000

Notes: Results are based on the semi-parametric ATE1 heterogeneous specifications of Tables 4 and 5, fit separately for target and control observations for each type of private equity buyout. Standard errors in parentheses. All reported coefficient values are statistically significant at the 0.01 level.

There are also good reasons to think employment effects vary by type of buyout. Public-to-private deals may be more likely to involve target firms with a strong
Table 7: effects by type of buyout

6.5 Tables 8–10: manufacturing productivity

Read:

- Table 8 shows that low-productivity target plants are more likely to exit, while new target plants are more likely to enter in the top tercile of the TFP distribution.
- Table 9 shows that entrants at target firms are much more productive than control entrants.
- Table 10 decomposes the estimated two-year manufacturing TFP gain of about 2.14 log points.
- Most of the TFP improvement comes from entry and exit rather than productivity growth at continuing establishments.

Economic message: Private equity raises productivity mainly through selection and reallocation: less productive plants exit, and more productive plants enter.

Table 8—ENTRY AND EXIT OF MANUFACTURING PLANTS BY LOCATION IN OWN-INDUSTRY DISTRIBUTION OF TOTAL FACTOR PRODUCTIVITY, BUYOUTS FROM 1980 TO 2003

<i>Panel A. Plant exit probabilities in the first two years post buyout (logistic specification)</i>			
Location in own-industry TFP distribution as of the buyout year t	Probability of plant exit by year $t + 2$		p -value for difference between targets and controls
	Targets	Controls	
Bottom tercile	0.143 (0.023)	0.091 (0.002)	0.025
Middle tercile	0.112 (0.034)	0.062 (0.002)	0.139
Top tercile	0.078 (0.015)	0.067 (0.002)	0.487
<i>Panel B. Plant entry probabilities in the first two years post buyout (logistic specification)</i>			
Location in own-industry TFP distribution in $t + 2$, two years after buyout	Probability that a plant operating in $t + 2$ entered in $t + 1$ or $t + 2$		p -value for difference between targets and controls
	Targets	Controls	
Bottom tercile	0.056 (0.015)	0.121 (0.006)	0.000
Middle tercile	0.071 (0.016)	0.078 (0.003)	0.590
Top tercile	0.127 (0.029)	0.072 (0.003)	0.058

Notes: Predicted probabilities implied by logistic regressions, using the propensity weights described in Section VIA. Standard errors are in parentheses. All reported coefficient values are statistically significant at the 0.01 level. The dependent variable in panel A equals 1 if the plant exits in the first or second year after the buyout year, zero otherwise. The dependent variable in panel B equals 1 if the establishment enters in the first or second year post buyout, zero otherwise. Entry and exit outcomes are determined using the full LBD. Explanatory variables are terciles of the own-industry $\times$ year TFP distribution interacted with target and control dummies. Terciles are defined based on a plant's position in the same-industry/same-year distribution of total factor productivity, using 4-digit SIC and digit NAICS industry definitions. The panel A regression has about 107,000 observations, and the panel B regression has about 91,000 observations.

Table 8: entry/exit by TFP tercile

Table 9—PRODUCTIVITY OF MANUFACTURING PLANTS, TARGETS AND CONTROLS, BUYOUTS FROM 1980 TO 2003

Panel A. TFP in buyout year t by plant status in year $t + 2$			p -value for difference between targets and controls
Plant status	Dependent variable: Plant-level log TFP in year t		
Plant status	Targets	Controls	
continuers	0.016 (0.012)	Omitted group	0.180
exits	-0.075** (0.035)	-0.032*** (0.008)	0.232
divestitures	-0.023*** (0.027)	-0.044*** (0.007)	0.462
χ^2	0.538		
Panel B. TFP in year $t + 2$, two years after buyout, by plant status in year $t + 2$			p -value for difference between targets and controls
Plant status	Dependent variable: Plant-level log TFP in year $t + 2$		
Plant status	Targets	Controls	
continuers	0.020* (0.011)	Omitted group	0.076
entrants	0.182*** (0.055)	-0.039*** (0.011)	0.000
acquisitions	-0.010 (0.047)	-0.030*** (0.007)	0.668
χ^2	0.523		
Panel C. TFP growth at continuing plants, from buyout year t to $t + 2$			p -value for difference between targets and controls
Plant status	Dependent variable: Change in plant-level log TFP from buyout year t to $t + 2$		
Plant status	Targets	Controls	
continuers	0.001 (0.011)	Omitted group	0.954
χ^2	0.071		

Notes: OLS regressions using the propensity weights described in Section VIA. The omitted group is continuing control plants. All specifications include industry-year effects as well as firm size and age effects. Standard errors are in parentheses. There are about 107,000 observations in the panel A regression, 91,000 in the panel B regression, and 62,000 in the panel C regression. On a propensity-weighted basis, 83 percent of the target observations in panel A are continuers, 8 percent are entrants, and 7 percent are divestitures. Continuers account for 86 percent of target observations in panel B, entrants for 8 percent, and acquisitions for 6 percent. Plant status is determined using the full LBD. Since target plants are overwhelmingly (99 percent) part of multi-unit firms, this analysis focuses on targets and controls part of multi-unit firms. By design, this table considers firms that continue from the buyout year to $t + 2$. If we add firm exits to the sample, the "Exits" row of panel A changes slightly: the coefficient becomes

Table 9: productivity of manufacturing plants

Table 10—IMPACT OF PRIVATE EQUITY BUYOUTS ON TOTAL FACTOR PRODUCTIVITY IN THE MANUFACTURING SECTOR, BUYOUTS FROM 1980 TO 2003

Estimated average two-year post-buyout change in TFP at target firms relative to controls, log points	
TFP log change differential	2.14
Excluding acquisitions and divestitures	2.01
Share of total TFP two-year change differential by margin of adjustment	
Continuing establishments	0.20
Entry and exit	0.74
Acquisitions and divestitures	0.06

Notes: Table entries are calculated according to equation (1) using difference in differences estimates from Table 9 and share measures retrieved from Table B.1 in the online Appendix, as discussed in the main text. The lower panel in the table reports the shares of the TFP log change differential on the left side of equation (1) accounted for by each term on the right side of the equation. The baseline average two-year TFP change for control firms is an estimated -0.38 log points with an estimated standard error of 0.24. This estimate is obtained from a propensity-weighted regression of the two-year log change in TFP on a constant and target status dummy in the sample of continuing target and control plants, the same sample used for panel C in Table 9.

Table 10: TFP decomposition

6.6 Table 11: earnings per worker

Read:

- Table 11 studies earnings per worker at target and control establishments.
- Target continuers have lower EPW growth relative to controls.
- Divestitures at target firms tend to involve high-EPW establishments.
- The overall EPW effect is negative for target firms relative to controls.

Economic message: Productivity gains are accompanied by reductions in earnings per worker, consistent with labour-cost reductions or compositional shifts toward lower-EPW establishments.

TABLE 11—EARNINGS PER WORKER (EPW) AT TARGET AND CONTROL ESTABLISHMENTS, BUYOUTS
FROM 1980 TO 2003

<i>Panel A. EPW in buyout year t by establishment status in year $t + 2$</i>			
Dependent variable: Establishment log real EPW in year t			p -values for difference between targets and controls
Establishment status	Targets	Controls	
Continuers	0.011 (0.003)	Omitted group	0.000
Exits	-0.085 (0.006)	-0.115 (0.004)	0.000
Acquisitions	0.163 (0.009)	-0.055 (0.006)	0.000
χ^2	0.448		
<i>Panel B. EPW in year $t + 2$, two years after buyout, by establishment status in year $t + 2$</i>			
Dependent variable: Establishment log real EPW in year $t + 2$			p -values for difference between targets and controls
Establishment status	Targets	Controls	
Continuers	-0.031 (0.003)	Omitted group	0.000
Entrants	0.015 (0.006)	-0.011 (0.004)	0.000
Acquisitions	0.010 (0.007)	-0.015 (0.006)	0.000
χ^2	0.421		
<i>Panel C. EPW growth at continuing establishments, from buyout year t to $t + 2$</i>			
Dependent variable: Change in establishment log real EPW from buyout year t to $t + 2$			p -values for difference between targets and controls
	Targets	Controls	
Continuers	-0.024 (0.002)	Omitted group	0.000
χ^2	0.200		

Table 11: earnings per worker

7 Detailed result synthesis

7.1 Employment effects

The establishment-level evidence shows modest relative job losses after buyouts. The effect is not immediate large-scale job destruction at all target establishments; rather, employment growth becomes weaker for target establishments over several post-buyout years. At the firm level, the net effect is moderated by new establishments and acquisitions.

7.2 Creative destruction

The paper's most important employment insight is that private equity buyouts raise both job creation and job destruction. This is why net employment effects are not sufficient. Private equity catalyzes reallocation across plants, business units, and establishments.

7.3 Productivity channel

In manufacturing, target firms improve TFP relative to controls. The decomposition suggests this improvement is mostly extensive-margin reallocation: closing low-productivity plants and adding higher-productivity plants. Continuing plant productivity growth plays a smaller role.

7.4 Worker earnings

The earnings-per-worker results show lower EPW at target establishments relative to controls. This suggests that private equity's productivity gains do not mechanically translate into higher average worker compensation, at least in the short window studied.

8 Short summary

- The paper studies 3,200 private equity buyouts and 150,000 target establishments from 1980–2005.

- It uses Census Bureau establishment and firm data to track employment, job flows, productivity, and earnings.
- Target establishments grow more slowly than matched controls after buyouts.
- Firm-level net job losses are modest because births and acquisitions offset losses among continuers and exits.
- Private equity buyouts strongly increase job creation, job destruction, job reallocation, and excess reallocation.
- Effects differ by deal type: public-to-private deals show stronger job losses, while independent-to-private deals can expand.
- Manufacturing target firms gain productivity relative to controls.
- TFP gains arise primarily through reallocation toward more productive plants.
- Earnings per worker decline at target establishments relative to controls.
- The overall lesson is that private equity buyouts accelerate creative destruction: they cut some jobs, create others, reallocate activity, and raise productivity.

9 Conclusion

9.1 Core conclusion

Davis et al. show that private equity buyouts have modest negative net employment effects but large effects on employment reallocation. Buyouts are associated with more job creation and more job destruction, and with productivity gains in manufacturing driven by reallocation across plants.

9.2 Economic intuition

Private equity owners restructure firms. This can mean closing weaker establishments, divesting units, acquiring new units, opening new establishments, and reallocating resources toward more productive operations. The resulting effect is not just downsizing. It is accelerated reallocation.

9.3 Takeaway

The paper's contribution is to replace a simple job-cutting narrative with a richer economic account. Private equity buyouts reduce some jobs and earnings, but also create jobs, shift activity across establishments, and raise manufacturing productivity. The welfare consequences therefore depend on the value of productivity gains, job reallocation, and distributional effects on workers.